

Assignment Title:

Understanding Fundamental Data Structures and Their Computational Concepts

Objective:

To connect the fundamental concepts of data structures with real-life applications and demonstrate understanding of how appropriate data structure selection improves system efficiency.

Task 1: Application Scenario

Instructions:

Choose one real-world system from the list below (or a similar one of your choice) and analyze how different data structures are used within it:

- Banking System (e.g., ATM Transaction Management)
- Navigation or GPS Application
- Hospital Management System
- Social Media Feed (e.g., Facebook/Twitter post management)
- Online Shopping Cart System
- Railway Reservation System

For your chosen system, do the following:

- Identify at least **three data structures** used (e.g., array, queue, linked list, tree, graph).
- Explain **why each structure is suitable** for the specific functionality it supports.
- Discuss **how improper data structure selection** could affect performance or resource usage.

Example (Sample Response Outline)

Chosen System: Online Shopping Cart

Functionality	Suitable Data Structure	Justification
Storing list of selected items	Array / List	Allows sequential storage and direct access for adding/removing products.
Managing checkout queue	Queue	Ensures first-come-first-served billing order.
Searching product by ID	Binary Search Tree (BST)	Enables fast search, insertion, and deletion operations.

Discussion on Improper Selection:

- Using a linked list instead of an array for product storage could slow down random access.
- Using a stack instead of a queue for checkout could process orders in reverse order, creating customer dissatisfaction.

Task 2: Reflection

Write a short reflection (150–200 words) addressing the following:

- What did you learn about the relationship between **data structure design and system performance**?
- How does understanding the **nature of data** help in selecting an efficient data structure?
- How can the same data structure behave differently depending on the **algorithm applied**?

Sample Reflection:

Understanding data structures helps optimize memory use and execution speed. In an e-commerce system, arrays simplify list management, while queues maintain order during transactions. Improper selection, such as using a stack for checkout, could reverse order processing and degrade user experience. This exercise demonstrated how theoretical concepts directly impact system efficiency and reliability, highlighting the importance of matching data structure properties to specific functionalities. It also showed that algorithm design and data structure choice are interdependent in achieving optimal performance.

Evaluation (5 Marks — Application & Reflection Component)

Criteria	Description	Marks
Application Accuracy	Correct identification and justification of data structures	2
Relevance to Real-Life Context	Logical link between data structure and system functionality	1
Reflection Quality	Depth of understanding and clarity of expression	2
Total		5

Guidelines

- 1. Complete all questions **individually** and neatly.
- 2. Include the following sections: **Cover Page, Objectives, Background, Algorithm/Flowchart (if applicable), Application & Reflection, Conclusion.**
- 3. Submit **typed or clearly handwritten work in PDF format.**
- 4. **Plagiarism or copied content will not be accepted.**
- 5. Follow professional formatting and **label all diagrams clearly.**
- 6. Submission file name format: **RollNo_Assignment1.pdf**
- 7. All assignments have been uploaded to the CR’s Google Drive (Folder: **CSE 1201 → Assignment 1**). Please share the folder link with the course teachers.
- 8. Deadline: **28 November 2025 (11:59 PM).**

Rubric (5 Marks) — Assignment: Understanding Fundamental Data Structures and Their Computational Concepts

Total Marks: 5

Criteria	Description	Marks
1. Application Accuracy	• Correctly identifies three data structures used in the chosen system • Provides clear, logical justification for why each data structure is suitable	2
2. Relevance to Real-Life Context	• Shows a meaningful connection between data structure properties and actual system functionality • Examples and explanation reflect real-world logic	1
3. Reflection Quality	• Reflection (150–200 words) demonstrates understanding of how data structure choice affects performance • Explains relationship between data, algorithms, and efficiency • Clear, coherent writing	2

Mapping Justification

Component	Mapping	Justification
CO	CO1	Understanding and applying linear & nonlinear data structures, stacks, queues, linked lists.
PO	PO1	Applying computer science knowledge to real-world system problems demonstrates engineering knowledge in practice.
Cognitive Level	C2 – Understand	Students interpret and explain the purpose and application of different data structures.
Knowledge Profile	K2	Requires conceptual understanding of formal computer science, algorithmic reasoning, and data modeling for analysis and selection.